Willian Vieira PhD

Geospatial machine learning engineer with a PhD in quantitative ecology. I build data systems that turn messy empirical data into reproducible simulations, transparent infrastructure, and decision-ready analytical products.

DATA SCIENCE & ENGINEER EXPERIENCE

2025 | 2024 (1 yr 9 m)

Data Analyst

Habitat, Montreal, Canada

Designed cloud data infrastructure on Azure for heterogeneous geospatial data, metadata-driven access, and feature retrieval for ML training and inference | Architected end-to-end R/Python ML pipelines to process and model large-scale, noisy real-world data for production analytics | Built metadata-driven ETL and provenance workflows to make datasets, models, assumptions, and analyses traceable, reproducible, and reliable | Led the transition to a Unix-based production environment using Docker, CI/CD, automated testing, and DevOps best practices | Pushed ML models into production, supporting client-facing data products with validation checks and monitoring

2024 | 2017

PhD Research

Integrative Ecology Lab, Sherbrooke, Canada

Built reproducible pipelines for continental-scale geospatial and environmental data, harmonizing climate, satellite, and field-inventory datasets across North America | Developed Bayesian hierarchical models and simulation workflows to represent heterogeneous forest populations, forecast dynamics, and propagate uncertainty from noisy, biased, incomplete real-world data | Ran thousands of simulations on HPC infrastructure () to test model behavior, compare assumptions, and scale analyses beyond local compute | Developed open-source software libraries (,) for spatial demographic modelling with version control, automated testing, and reusable workflows | Published a technical methods book documenting the full computational pipeline, along with one peer-reviewed publication and two preprints (,)

2022 | 2020 (part-time)

Biostatistician

Environment and Climate Change Canada - Quebec, Canada

Developed a cost-aware probability sampling protocol to improve the spatial representativeness of boreal bird surveys in Quebec | Led R&D on spatial bias-correction methods for large-scale ecological monitoring data, designing an approach later adopted by other provinces | Engineered a fully automated and reproducible data pipeline with version-controlled workflows, automated documentation, and stakeholder-ready analytical outputs

EDUCATION

2024 | 2017

PhD, Ecology

Université de Sherbrooke - Sherbrooke, Canada

How climate, competition, and forest management shape the limits of tree species distributions: from individuals to metapopulations

2016 | 2015

Masters 2, Agroecology and Resource Management

Bordeaux Sciences Agro, Bordeaux, France

Modelling the dispersion of weed species in agricultural landscapes

2015 | 2010

BSc in Agronomy

Universidade Federal de Santa Catarina, Florianópolis, Brazil

CONTACT

(263) 381-3636
w.vieiraw@gmail.com
Linkedin/WillVieira
GitHub/WillVieira
Publications

SKILLS

ML & Statistics: Bayesian inference · Hierarchical models · Simulation modelling · Spatial modelling · Uncertainty quantification · Model evaluation

Programming & Database: Python · R · SQL · Bash · DuckDB · Stan · Julia

Data Engineering & Cloud: ETL/ELT · Geospatial data products · Cloud data infrastructure · APIs · Docker · Azure · AWS · GeoParquet · COGs · Apache Arrow · HPC

DevOps & Reproducibility: Git · CI/CD · Unix · Unit testing · Reproducible pipelines · Automated validation · Metadata · Provenance · Make

OUTREACH

· Developed and maintained the QCBS R Workshop Series, reaching nearly one thousand graduate students · Taught programming, statistics, and ecology to 250+ students across biology, engineering, and graduate programs · Published 5 papers & preprints with ~100 citations · Translated stakeholder needs into technical analyses, reproducible workflows, and clear analytical outputs

LANGUAGES

Portuguese · Native
French · Full Professional
English · Full Professional

Source · HTML · PDF
Updated 2026-05-27